Foraging Behaviour Model

How does an animal learn where to forage?

A reinforcement learning simulation in which an animal chooses between food patches while balancing food gain and movement cost.

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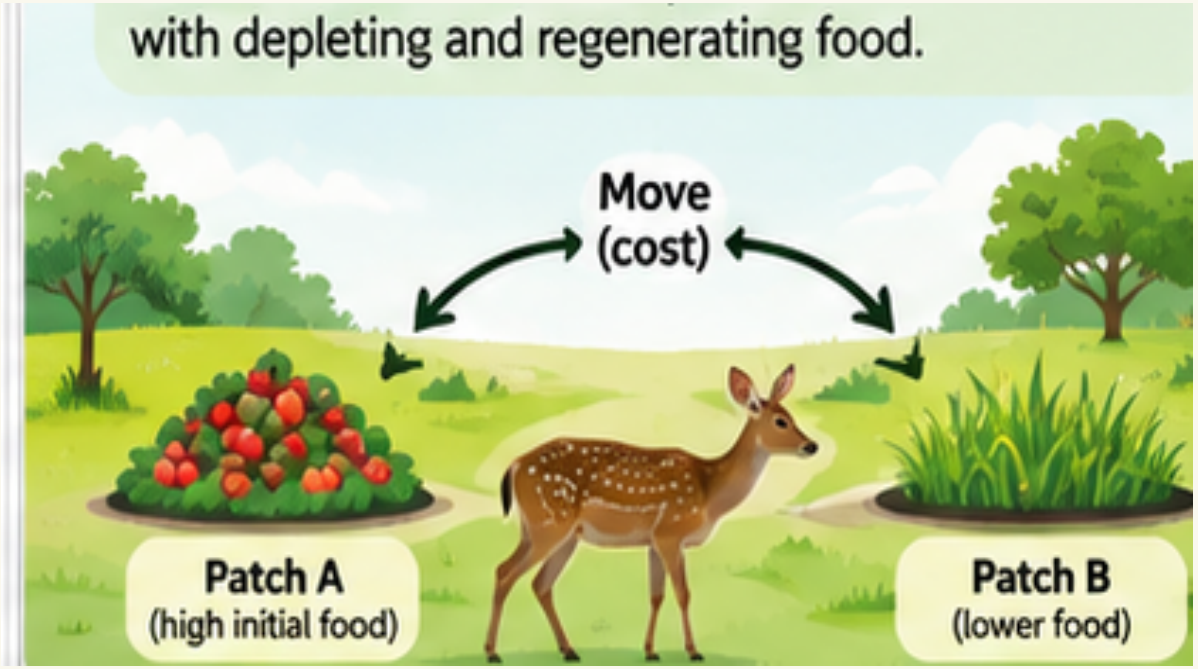
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PROJECT SNAPSHOT

- Agent: simulated animal
- Method: tabular Q-learning
- Evaluation: 3 decision strategies

Environment & RL Formulation



Patch A: capacity 10 units | Patch B: capacity 8 units

Forage: collect up to 2 units | Regeneration: +1 per step

MARKOV DECISION PROCESS

STATE	Current patch + food levels in A and B
ACTION	Forage here | Move to other patch
REWARD	Food collected | –0.5 for movement
GOAL	Maximize cumulative reward over 60 steps

Two actions • Discrete state space • 60-step episodes

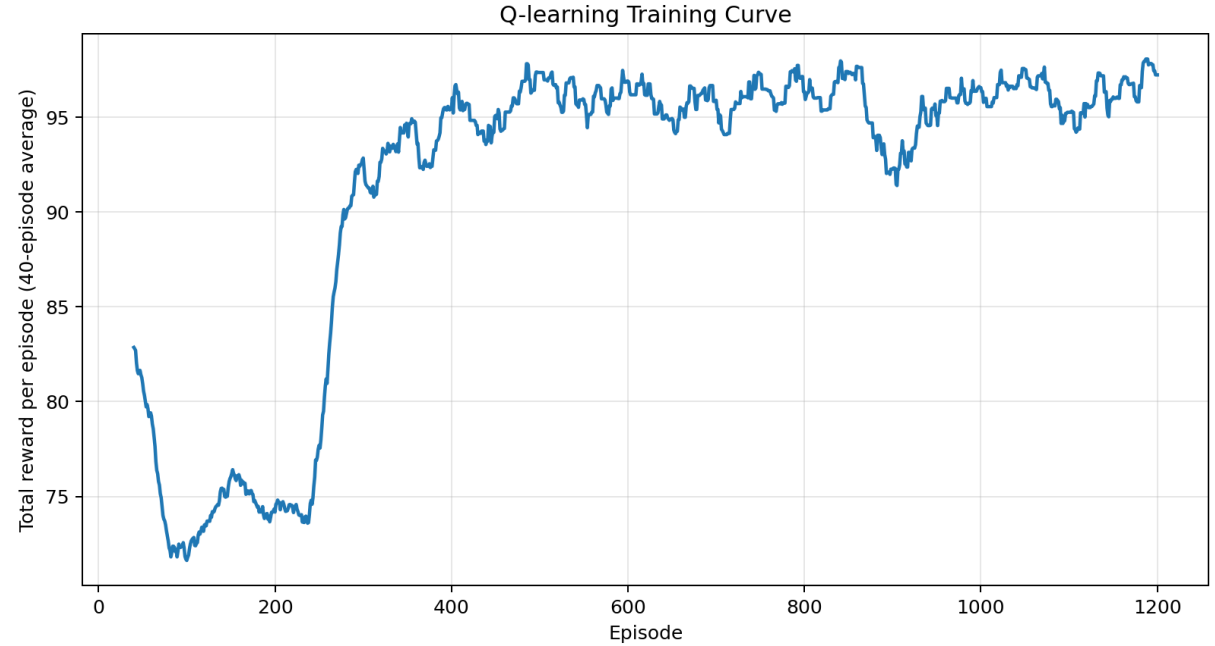
Learning Method: Tabular Q-learning

Q-LEARNING UPDATE

$$Q(s,a) \leftarrow Q(s,a) + \alpha\,[r + \gamma \max Q(s',a') - Q(s,a)]$$

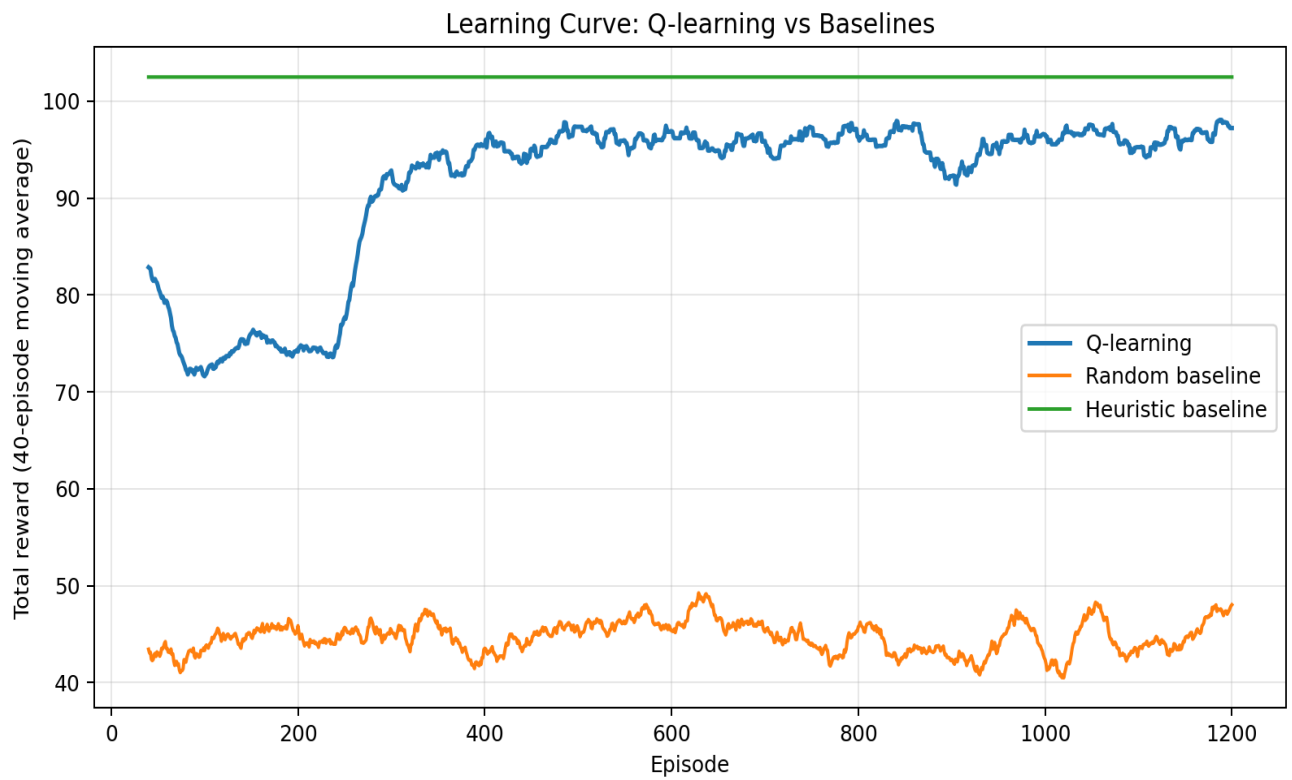
- α = 0.15 | learning rate
- γ = 0.95 | discount factor
- ε starts at 0.20 and decays
- 1,200 training episodes

TRAINING REWARD



40-episode moving average; final value: 97.24

Experimental Results & Comparison



Mean cumulative reward over 200 evaluation episodes (± standard deviation)

MEASURED MEAN REWARD

Q-learning	100.00
Random	45.28
Heuristic	102.50

Finding: Q-learning beats random; the simple heuristic scores slightly higher.

Discussion, Ethics & Conclusion

LIMITATIONS

- Only two food patches and two actions.
- Food regeneration is deterministic.
- Agent observes exact food levels.
- No predators, weather or competition.

ETHICS & TAKEAWAY

- Simulation only: no live animals or personal data.
- Use biological models cautiously; real ecosystems are more complex.
- Q-learning learns a useful policy and clearly improves on random choice.
- Future work: uncertain observations, more patches and variable food growth.



Thank you